

Slide 1: Defining Functional Dependencies

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$X \rightarrow Y$ means the value of attribute set X uniquely determines the value of attribute set Y in every valid relation state.

Slide 2: Armstrong's Axioms

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Reflexivity (if $Y \subseteq X$, $X \rightarrow Y$), Augmentation (if $X \rightarrow Y$, $XZ \rightarrow YZ$), Transitivity (if $X \rightarrow Y$ and $Y \rightarrow Z$, $X \rightarrow Z$).

Slide 3: Attribute Closure Algorithm

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Computing X^+ under a set of FDs to determine candidate keys and superkeys.